

1. Create an “**Empty Assembly-only Project**” in Code Composer Studio for **Task 1** and **Task 2** separately.
2. Update **main.asm** template assembly file so as to complete each task below and rename the file name as **main1.asm** and **main2.asm** accordingly.
3. **Task 1**
 - **main1.asm**
 - Write an assembly program that corresponds to the following C language program, where
 - i. Create two random integer lists (**list1** and **list2**), each with 10 **word** elements in **.data** section,
 - ii. Create a **word** variable (**result**) and initialize it 0 in **.data** section,
 - iii. All **pointer** arguments of **sublab1** subroutine must be passed using stack, and the others by means of an argument register(s).
 - iv. In **sublab1** subroutine, you are allowed to use and modify only those **callee saved registers** (i.e., push them into stack before use).

```
int list1[10], list2[10];
int result;

void sublab1(int *ls1, int *ls2, int N, int *res)
{
    int i, sum=0;

    for(i=0; i<N; i++) {
        sum += ls1[i]+ls2[i];
        ls1[i]=ls2[N-1-i];
    }

    *res=sum;
}

int main() {
    sublab1(list1, list2, 10, &result);
}
```

4. Task 2

- **main2.asm**
- Write an assembly program that corresponds to the following C language program, where
 - i. Create a random integer lists (**list1**), with 10 **word** elements in **.data** section,
 - ii. All **pointer** arguments of **sublab2** subroutine must be passed using stack, and the others by means of registers.
 - iii. If you use any **callee saved registers** in a subroutine, you need to push them into stack before use.
 - iv. You need to create two local variables **a1** and **b1**.

```
int list1[10];

int sub (int *a, int *b) {
    return (*a- *b);
}

int compare (int a, int b) {
    int a1=a*2;
    int b1=b+1;
    if (sub(&a1, &b1) >= 0) return 1;
    else return 0;
}

void sublab2(int *ls, int N, int M) {
    for(int i=0;i<N;i++)
        ls[i]=compare(i,M+i);
}

int main() {
    sublab2(list1, 10, 6);
}
```